

BUTAVENT RESPIRATOR SOLUTION

DESCRIPTION: A clear, colourless solution to pale yellow solution.

COMPOSITION: Each ml contains Salbutamol Sulphate BP equivalent to Salbutamol 5.0 mg

Preservative: Benzalkonium Chloride 0.04mg/ml

PHARMACODYNAMICS: The pharmacologic effects of beta-adrenergic agonist drugs, including albuterol, are at least in part attributable to stimulation through beta- adrenergic receptors of intracellular adenylyl cyclase, the enzyme that catalyzes the conversion of adenosine triphosphate (ATP) to cyclic -3', 5'-adenosine mono-phosphate (cyclic AMP). Increased cyclic AMP levels are associated with relaxation of bronchial smooth muscle and inhibition of release of mediators of immediate hypersensitivity from cells, especially from mast cells.

Salbutamol is a relatively selective beta-2 adrenoceptor stimulant. It is more specific than both isoprenaline and orciprenaline for adrenergic beta 2-receptors. After oral and parenteral administration, stimulation of the beta receptors in the body, both beta-1 and beta-2, occurs because (a) beta-2 selectively is not absolute, and (b) higher concentrations of salbutamol occur in the regions of these receptors with these modes of administration. This results in the beta-1 effect of cardiac stimulation, though not so much as with isoprenaline, and beta-2 effects of peripheral vasodilatation and hypotension, skeletal muscle tremor and uterine muscle relaxation.

Metabolic effects such as hyperinsulinaemia and hyperglycaemia also may occur, although it is not known whether these effects are mediated by beta-1 or beta-2 receptors. The serum potassium levels have a tendency to fall.

PHARMACOKINETICS: Following inhalation salbutamol acts topically on bronchial smooth muscle and initially the drug is undetectable in the blood. After 2 to 3 hours low concentrations are seen, due presumably to the portion of the dose which is swallowed and absorbed in the gut.

The elimination half-life of inhaled or oral salbutamol is between 2.7 and 5 hours. Salbutamol is not metabolised in the lung but is converted to the 4'-o-sulphate ester in the liver. Salbutamol is excreted in the urine as free drug and as the metabolite. After oral administration 58-78% of the dose is excreted in the urine in 24 hours, approximately 60% as metabolites. A small fraction is excreted in the faeces. Impairment of liver or renal function may necessitate a reduction in dosage.

INDICATION: For the treatment of acute severe asthma (status asthmaticus) and for routine management of chronic bronchospasm unresponsive to conventional therapy.

RECOMMENDED DOSAGE: To avoid cross contamination, proper aseptic technique should be used.

The usual dosage for adults and children 12 years and older is 0.5 mL of salbutamol administered three to four times daily by nebulization. More frequent administration or higher doses are not recommended. To administer 0.5 mL of salbutamol, dilute 0.5 mL of the 0.5% inhalation solution with 2.5 mL of sterile normal saline solution. The flow rate is regulated to suit the particular nebulizer so that salbutamol sulphate inhalation solution will be delivered over approximately 5 to 15 minutes.

Children under 12 years of age: The usual dosage is 0.5ml (2.5mg Salbutamol) diluted to 2.0 or 2.5ml using normal saline for injection as diluent. Some children may however require higher doses of salbutamol up to 5.0 mg. Efficacy of nebulised salbutamol in infants under 18 months is uncertain. As transient hypoxaemia may occur, supplemental oxygen therapy should be considered.

The use of salbutamol sulphate inhalation solution can be continued as medically indicated to control recurring bouts of bronchospasm. During this time most patients gain optimal benefit from regular use of the inhalation solution.

If a previously effective dosage regimen fails to provide the usual relief, medical advice should be sought immediately as this is often a sign of seriously worsening asthma that would require reassessment of therapy.

Drug compatibility (physical and chemical), efficacy, and safety of salbutamol sulphate inhalation solution, when mixed with other drugs in a nebulizer have not been established.

ROUTE OF ADMINISTRATION: Nebulization

CONTRAINDICATIONS: Hypersensitivity to any of the ingredients.

WARNINGS & PRECAUTIONS: The management of asthma should normally follow a stepwise programme, and patient response should be monitored clinically and by lung function tests. Increasing use of short acting inhaled beta-2 agonists to control symptoms indicates deterioration of asthma control. Under these conditions, the patient's therapy plan should be reassessed. Sudden and progressive deterioration of asthma control is potentially life-threatening and consideration should be given to starting or ceasing corticosteroid therapy. In patients considered at risk, daily peak flow monitoring may be instituted. Patients should be warned that if either the usual relief is diminished or the usual duration of action reduced, they should seek medical advice at the earliest opportunity after increasing the dose.

Animal studies suggest that cardioneurotic effects may occur with high dosages of some sympathomimetic amines. On this evidence the possibility of the occurrence of myocardial lesions cannot be excluded subsequent to long term treatment with these drugs.

Care should be taken with patients who are known to have received large doses of salbutamol or other sympathomimetic drugs, or who are suffering from hypertension, hyperthyroidism, cardiac insufficiency, or diabetes mellitus. Salbutamol should be administered cautiously to patients with thyrotoxicosis.

Excessive use may induce a non-responsive state leading to a worsening of hypoxaemia.

Patients receiving treatment at home must be warned that if either the usual relief is diminished or the usual duration of action reduced, they should not increase the dose or its frequency of administration but should seek medical advice.

Potentially serious hypokalaemia may result from beta2-agonist therapy mainly from parenteral and nebulised administration. Particular caution is advised in severe asthma as this effect may be potentiated by concomitant treatment with xanthines derivatives, steroids, diuretics and by hypoxia. It is recommended that serum levels are monitored in such situations.

A small number of cases of acute angle closure glaucoma have been reported in patients treated with a combination of nebulised salbutamol and ipratropium bromide. A combination of nebulized salbutamol with nebulised anticholinergics should therefore be used cautiously. Patients should receive adequate instruction in correct administration and be warned not to let the solution or mist enter the eye. Due to the blood glucose increasing effects of beta2-stimulants, extra blood glucose controls are initially recommended when diabetic patients are commenced on salbutamol.

INTERACTION WITH OTHER MEDICAMENTS: Beta adrenergic blocking drugs inhibit the bronchodilator action of salbutamol and other sympathomimetic bronchodilators. However, such drugs should not be used in asthmatic patients as they may increase airway resistance.

Care is recommended if it is proposed to administer salbutamol in concomitant therapy with other sympathomimetic amines as excess sympathetic stimulation may occur.

Animal studies have shown that large doses of salbutamol may interact with imipramine, chlorthalidone and chlorpromazine but any practical significance of these results in man remains to be established. Concurrent use of salbutamol with diuretics, xanthines or high doses of corticosteroids may increase the risk of hypokalaemia. The antihypertensive effects of diuretics may be reduced when these medications are with used salbutamol.

USE IN PREGNANCY & LACTATION:

Use in pregnancy: Salbutamol is known to cross the placental barrier in humans. Safety for use in pregnancy has not been demonstrated, therefore the drug should not be used in pregnant women, or those likely to become pregnant, unless the expected benefit outweighs any potential risk.

Use in lactation: It is not known whether salbutamol is excreted in breast milk or whether it has a harmful effect on the newborn. Therefore, it is not recommended for nursing mothers unless the expected benefits outweigh any potential risk.

SIDE EFFECTS: A fine tremor of skeletal muscle has been reported in some patients when salbutamol is administered orally or by inhalation, the hands being the most obviously affected; a few patients feel tense. These effects are dose related and are caused by a direct action on skeletal muscle and not by direct CNS stimulation.

Increases in heart rate may occur in patients with normal heart rate after administration of salbutamol respirator solution. These increases are dose dependent and are of the order of 9 beats/minute when 10 mg of salbutamol as 0.5% w/v solution is inhaled by adults over 3 minutes, 13 beats/minute when 20 mg of salbutamol as 0.1% w/v solution is inhaled by adults over 3 minutes. In patients with pre-existing sinus tachycardia, especially those in status asthmaticus, the heart rate tends to fall after the administration of salbutamol respirator solution as the condition of the patient improves.

With higher doses than those recommended, or in patients who are unusually sensitive to beta-adrenergic stimulants, dilatation of some peripheral arterioles may occur leading to a small reduction in arterial pressure; a compensatory increase in cardiac output may then occur.

Other reactions which may occur are headaches, nausea, palpitations and sensations of warmth. Hypersensitivity reactions including angioedema, urticaria, bronchospasm, hypotension and collapse have been reported very rarely. There have been very rare reports of muscle cramps. Mouth and throat irritation may occur with inhaled salbutamol.

Note: The incidence and severity of particular side effects depends on the dosage and route of administration. Salbutamol sulphate does not cause difficulty in micturition because, unlike sympathomimetic drugs such as ephedrine, therapeutic doses have no alpha-adrenergic receptor stimulant activity.

Potentially serious hypokalaemia may result from beta-2-agonist therapy.

As with other inhalation therapy the potential for paradoxical bronchospasm should be kept in mind. If it occurs, the preparation should be discontinued immediately and alternative therapy instituted. As with other beta-2 agonists, hyperactivity has been reported rarely in children.

SYMPTOMS AND TREATMENT OF OVERDOSE: The signs of salbutamol overdosage are significant tachycardia and/or significant muscle tremor.

The specific antidote for over dosage with salbutamol is a cardioselective beta-blocking agent given by intravenous injection. In general, beta-blocking drugs should be used with caution as they may cause bronchospasm in sensitive individuals.

Hypokalaemia may occur following overdosage with salbutamol. Serum potassium levels should be monitored.

STORAGE CONDITIONS: Store below 30°C. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

Any solution remaining in the bottle once it has been opened should not be used later than 1 month

SHELF LIFE: Please refer to outer package.

PACK SIZE: Pack in a carton of 25 x 10 ml glass amber bottles

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

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